

Project Report Finding Similar Items

Francesco Memmola

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Master in Data Science for Economics
Università degli Studi di Milano

Project Repository: https://github.com/FraSeb6/Similarity_Ish_Abstract

1 Introduction

The goal of this project is to detect pairs of highly similar abstracts inside the arXiv corpus (roughly 3 million scientific papers). The difficulty is not conceptual but computational: the similarity measure of interest is the **Jaccard similarity** between abstracts, and comparing every possible pair would cost $O(n^2)$ — about 4.5×10^{12} comparisons on the full corpus, which is computationally infeasible.

The classical **Shingling** $\rightarrow$ **MinHash** $\rightarrow$ **LSH** pipeline is therefore adopted, estimating Jaccard similarity through compact signatures and restricting comparisons to document pairs that are already likely to be similar. Two design priorities guided every choice:

- Each step is implemented to reflect its theoretical definition as closely as possible, making the connection between the algorithm and its mathematical foundation explicit and verifiable.
- Both **precision** and **recall** are measured against an exact ground truth, and the resulting pairs are grouped into **duplicate clusters**. As shown in Section 3, measuring recall is precisely what allowed the diagnosis and correction of a poor parameter choice.

A single global flag, `USE_SUBSAMPLE`, switches the same code between a fast subsample (used for the numbers reported here) and the complete dataset.

2 Data Ingestion and Cleaning

The arXiv metadata snapshot (`Cornell-University/arxiv`) is downloaded through the Kaggle API as a single ~ 4 GB file with one JSON object per line.

Streaming. The file is never loaded into memory as a whole. It is read *line by line*: each record is parsed, only the **abstract** field is kept, and it is appended to a growing list. Memory therefore depends only on how many abstracts are kept, not on the file size.

Quality filter. Very short abstracts — placeholders and “this paper has been withdrawn” notices — produce tiny shingle sets that generate spurious high-similarity matches. Any abstract shorter than 20 words is discarded directly inside the streaming loop. On the 50,000-line working slice this removes 673 entries, leaving **49,327** clean abstracts, which are then collected into a `pandas DataFrame`.

Normalisation. Before shingling, each abstract is lower-cased and all runs of whitespace (spaces, tabs, newlines) are collapsed to a single space. Normalisation is kept deliberately light: removing formatting noise is enough, while keeping the procedure simple to justify.

3 From Abstracts to Sets: Character Shingling

Each abstract is turned into the **set of its k -character shingles** — every substring of k consecutive characters. Two abstracts that share many shingles are similar, so the Jaccard similarity of these sets is exactly the quantity of interest.

3.1 Choosing k by experiment

The value of k is the single most consequential parameter, and it was fixed empirically through experimentation.

With $k = 5$ the shingles are too generic: fragments such as “the”, “tion” or “ing” occur in almost every English abstract, producing a large number of false positives — roughly 50,000 candidate pairs and a precision of only a few percent.

With $k = 9$ shingles are rare enough to be meaningful: the candidate set shrank to a few hundred pairs, precision rose to about 63%, and recall stayed high. $k = 9$ is therefore adopted.

3.2 Hashing the shingles

Storing millions of string shingles would be memory-intensive, so each shingle is hashed to an integer using Python’s `zlib.crc32`. The unique integers are stored in a Python `set`, so each shingle is counted once.

4 MinHash Signatures

Comparing full shingle sets pairwise is computationally infeasible, so MinHash compresses each set into a fixed-length **signature** of $M = 100$ integers.

4.1 Principle

Given a random permutation h of the shingle universe, the probability that two sets share the same minimum element equals their Jaccard similarity:

$$P(\min h(A) = \min h(B)) = J(A, B). \quad (1)$$

Using M independent permutations, the fraction of matching signature positions estimates J . Permutations are simulated through a family of linear hash functions:

$$h_{a,b}(x) = (a \cdot x + b) \bmod p, \quad (2)$$

where x is an integer representing a shingle, p is a prime number, and a, b are random coefficients generated with a fixed seed to ensure the reproducibility of experiments.

4.2 Implementation

The signature of each document is initialised with all values set to $+\infty$. For each shingle x , all 100 hash functions are evaluated at once using NumPy, updating the signature with the element-wise minimum:

```
sig = np.minimum(sig, (A * x + B) % PRIME)
```

This is the direct definition of MinHash applied to the document’s shingles. Signatures are stored in an in-memory integer matrix of shape $N \times 100$.

5 Locality-Sensitive Hashing with a Derived Split

Even with compact signatures, comparing all of them is still $O(n^2)$. LSH avoids this by hashing similar signatures into shared buckets.

5.1 Banding

The $N \times 100$ signature matrix is split into b bands of r rows each, with $b \cdot r = 100$. Two documents that are identical in at least one band fall into the same bucket and become a **candidate pair**. For a pair with true similarity s , the probability of becoming a candidate follows the S-curve:

$$P(\text{candidate}) = 1 - (1 - s^r)^b, \quad (3)$$

whose steep point sits near the threshold:

$$t \approx \left(\frac{1}{b}\right)^{1/r}. \quad (4)$$

5.2 Deriving b and r

Instead of hard-coding b and r , they are derived from a target threshold. Given `TARGET_THRESHOLD` = 0.5, the code enumerates every factorisation of 100 into $b \cdot r$ and keeps the one whose threshold is closest to the target. This selects $b = 20$ **bands** of $r = 5$ **rows**, with:

$$t = \left(\frac{1}{20}\right)^{1/5} \approx 0.549. \quad (5)$$

The resulting S-curve is shown in Figure 1.

5.3 Bucketing

For each band, the 5-integer slice of each document is converted to a byte string using `tobytes()`, and used as a dictionary key to group documents that share it. For every bucket with at least two documents, `itertools.combinations` generates the candidate pairs. On the working slice this produces **380 candidate pairs** in about one second.

6 Evaluation: Precision and Recall

LSH is approximate, so its output must be validated. Both the fraction of returned pairs that are correct (**precision**) and the fraction of true pairs that were found (**recall**) are measured — precision alone cannot reveal the pairs that were missed.

6.1 Exact Ground Truth

On a small subsample of 4,000 documents the *exact* Jaccard similarity of *all* pairs is computed by brute force. This is $O(n^2)$ and is only affordable on a small slice, but it gives the true set of similar pairs: at threshold 0.549 there are **7** of them.

6.2 Results

Restricting the candidates to the same 4,000-document slice, LSH recovers 6 of the 7 true pairs, i.e. a **recall of 85.7%**. Since this slice contains only a handful of candidates, a more robust **global precision** is also computed over all 380 candidate pairs, recomputing their exact Jaccard: 63.2% of them are genuinely similar. Figure ?? shows the distribution of exact similarities for the full candidate set, with a clear mass of true duplicates above the threshold and a tail of false positives below it.

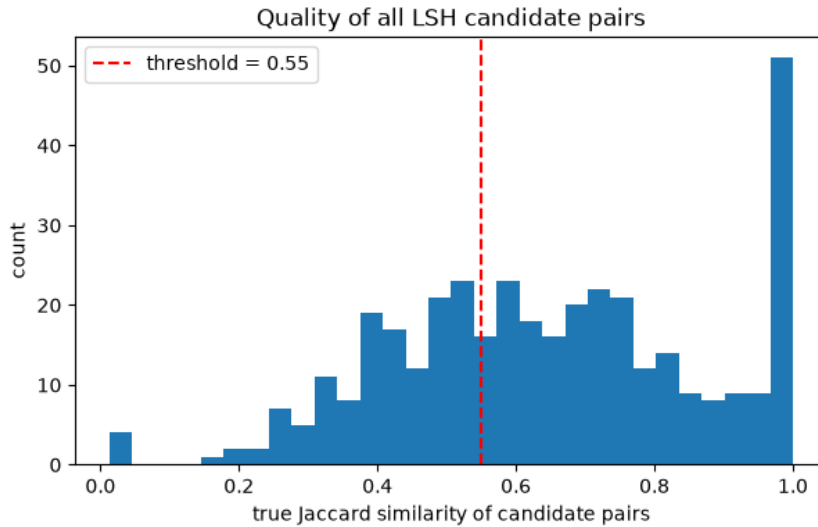


Figure 1: Exact Jaccard similarity of all 380 candidate pairs. The dashed line is the LSH threshold; candidates to its right are true near-duplicates.

7 Duplicate Clusters

The candidate pairs are used to build a graph in which each abstract is a node and each pair is an edge. Groups of mutually similar abstracts correspond to the **connected components** of this graph, computed through a depth-first traversal implemented directly without external libraries. On the working slice several duplicate groups are identified. The largest contains a series of papers sharing almost identical text — a confirmation that the pipeline isolates real duplicates rather than random matches.

8 Scalability

To show the pipeline scales, we time MinHash and LSH on growing prefixes of the data and append, as a final point, the run on the whole loaded dataset — reusing the times already measured in the main pipeline, so this last row costs nothing extra. Results are in Table 1 and Figure ??.

Both stages grow **linearly** in n : total time roughly doubles when n doubles, confirming we have bypassed the $O(n^2)$ barrier. MinHash dominates; LSH is negligible. Extrapolating the

Documents (n)	MinHash	LSH	Total	Candidates
5,000	5.3 s	0.0 s	5.4 s	15
10,000	10.9 s	0.2 s	11.1 s	35
20,000	21.6 s	0.4 s	22.0 s	88
40,000	42.8 s	0.7 s	43.5 s	298
49,327	51.2 s	1.1 s	52.3 s	380

Table 1: Execution time per stage on increasing dataset sizes. Time grows linearly with n .

slope, the full $\sim 3\text{M}$ -abstract corpus is processed in on the order of 50 minutes for MinHash, by simply setting `USE_SUBSAMPLE = False` — the same code, unchanged.

Limitations. The metrics above are measured on a 50,000-abstract subsample; recall in particular is estimated from a small ground truth, so it carries sampling noise. Finally, recall below 100% is the expected price of LSH: a small fraction of true pairs can be missed when they never collide in any band.

9 Conclusion

A complete pipeline for near-duplicate detection on arXiv abstracts has been implemented using Shingling, MinHash, and LSH over the Jaccard similarity. A dual precision/recall evaluation and a clustering step turned the raw candidate pairs into interpretable, validated results. The empirical choice of 9-character shingles and the data-driven 20×5 banding illustrate that careful parameter selection, guided by measurement, matters as much as the algorithm itself. The pipeline scales linearly and runs on the full dataset through a single flag.

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